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Ionization rates and critical fields in 4H silicon carbide

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Epitaxial p-n diodes in 4H SiC are fabricated showing a good uniformity of avalanche multiplication and breakdown. Peripheral breakdown is overcome using the positive angle beveling technique. Photomultiplication measurements were performed to determine electron and hole ionization rates. For the electric field parallel to the c-axis impact ionization is strongly dominated by holes. A hole to electron ionization coefficient ratio of up to 50 is observed. It is attributed to the discontinuity of the conduction band of 4H SiC for the direction along the **c** axis. Theoretical values of critical fields and breakdown voltages in 4H SiC are calculated using the ionization rates obtained. © *1997 American Institute of Physics.* [S0003-6951(97)03427-X]

Many important device applications of silicon carbide are based on its unusually high electric strength. Breakdown fields in hexagonal silicon carbide are nearly ten times higher than in silicon which means a possibility to considerably increase the operation speed of high power and high voltage devices[1–3]. Besides, the information on avalanche breakdown in p-n junction devices in SiC is very limited, particularly for the 4H polytype, the SiC polytype with a high electron mobility. The major problem comes from immaturity of material technology: real devices typically have an early breakdown either via the device periphery or at crystal imperfections. This makes it difficult to obtain conclusive data on critical fields and ionization rates.

In the present letter we report on the first epitaxial 4H SiC diodes with uniform avalanche multiplication and breakdown. Electron and hole ionization rates are measured using the photomultiplication technique. Theoretical dependencies for critical fields and breakdown voltages in 4H SiC are calculated using the experimental values of ionization rates.

p-n diodes were fabricated from p^+-n-n^+ epitaxial structures. Two types of epitaxial techniques were used for diode fabrication: sublimation epitaxy and vapor phase epitaxy (VPE). The low voltage diode (sample No. 1) was grown by sublimation epitaxy and processed using the technique previously described in Ref. 4. The diode has a breakdown voltage of about 64 V and a slightly graded doping profile, with a base doping of $0.7–1.2\times10^{18}$ cm^{-3}. p-n diodes with higher breakdown voltages (sample Nos. 2 and 3) were grown using the VPE technique[5], depositing n^0 and n^+ layers onto a p^+ substrate. The doping profiles of VPE-grown diodes are abrupt as was established by capacitance-voltage (C-V) measurements. The base doping was 3.4×10^{17} cm^{-3} for sample Nos. 2 and 5.2×10^{16} cm^{-3} for sample No. 3. The base n^0 layer was in each case thick enough to prevent the electric field reach through. The p-n junctions were oriented in the (0001) crystal plane with an off-orientation angle of 3.5° or lower.

VPE-grown structures were processed so as to form positive angle-beveled mesa diodes with an angle of about 45°, as is shown in the inset in Fig. 1. p-n diodes with a positive angle bevel have a lower electric field at the device periphery than in the bulk[6]. Avalanche breakdown in such diodes is controlled by bulk material properties rather than by peripheral effects. The diode mesas used for measurements were squares, 0.2×0.2 mm large.

Reverse-bias current voltage (I-V) characteristics and photomultiplication were studied using Keithley Model 237 Source Measure Unit. Sample Nos. 1 and 2 have low breakdown voltages and could be characterized in air without problems. Measurements on sample No. 3 were done in Fluorinert (a high dielectric strength liquid) to avoid air spark formation. It is essential for investigations of avalanche breakdown to assure a good breakdown homogeneity. A simple but efficient technique to evaluate the homogeneity of avalanche breakdown in SiC is the observation of breakdown luminescence. A comparison between images of the breakdown luminescence and of electron beam induced current shows a very close correlation for 6H SiC[4]. In the present study we also observed a correlation between the bright spots of light emission and microplasma kinks of the reverse-bias I-V curve. Only the diode mesas free from microplasma effects were used for photomultiplication studies.

Photomultiplication was measured in the dc mode using a UV He–Cd laser ($\lambda=325$ nm). This is a penetrating light for the p^+-n junctions used in the present study. The photo-

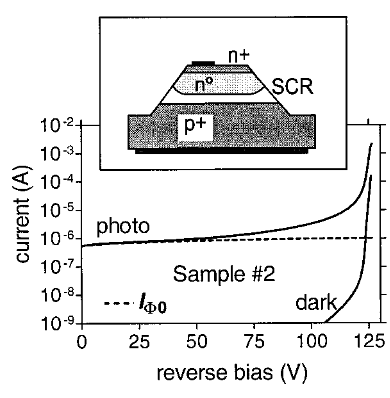

FIG. 1. Dependencies of photo and dark currents on reverse bias voltage for sample No. 2. The dashed line shows the initial photocurrent, $I_{\Phi0}$. The inset shows the configuration of the VPE-grown test diodes with positive angle beveling.

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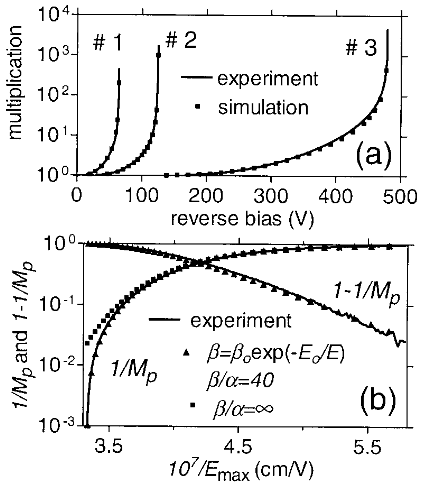

FIG. 2. (a) Dependencies of the multiplication coefficient M on reverse bias voltage for samples used in the study. Squares show results of the simulation using the extracted values of ionization rates. (b) The procedure of fitting to photomultiplication data: dependencies of $1/M$ and of $1-1/M$ vs reciprocal maximum field in sample No. 3 (lines). Simulated curves (points) use $E_o = 1.67\times10^7$ V/cm; $\beta_0 = 1.63\times10^7$ cm^{-1}, $\beta/\alpha = 40$ and $\beta/\alpha = \infty$.

current I_Φ is generally $I_\Phi = I_{\Phi 0}M$, where M is the multiplication coefficient, $I_{\Phi 0}$ the initial photocurrent. For the voltage range between $(1/2)V_b$ and V_b (V_b being the breakdown voltage) the photocurrent increases by three to four orders of magnitude, as is seen in Fig. 1. The increase is largely due to avalanche multiplication. However, an appropriate correction must be done to exclude the contribution from the increase of the initial photocurrent, i.e., from $I_{\Phi 0}$[4]. Plots of multiplication coefficients versus reverse bias for the samples Nos. 1–3 are shown in Fig. 2(a). Very high multiplication coefficients can be reached, up to 5000.

The multiplication coefficient is generally given by $M = f_n M_n + f_p M_p$, M_n and M_p being the coefficients of electron-initiated and hole-initiated multiplication, f_n and f_p the electron and hole fractions of the initial photocurrent. For an asymmetric p^+-n diode the initial photocurrent is dominated by the photogenerated holes coming from the n^0 region, $f_n \ll f_p$. The multiplication coefficient therefore corresponds to hole-initiated multiplication, $M \approx M_p$. It is seen from Fig. 2(a) that the hole-initiated multiplication in sample No. 3 starts at a voltage below $(1/2)V_b$ and a high multiplication is observed for voltages noticeably below V_b. Certain conclusions on the type of multiplication process can immediately be made from the shape of photomultiplication curves. It is seen from Fig. 2(a) that the hole-initiated multiplication in sample No. 3 starts at a voltage below $(1/2)V_b$ and a high multiplication is observed for voltages noticeably below V_b. This type of behavior can only be observed if the photocarriers (holes in our case) dominate the charge multiplication process, $\beta \gg \alpha$, where β and α are the hole and electron ionization rates, respectively. For $\beta \ll \alpha$ the hole initiated multiplication is only observed at $V > 0.7$–$0.9\ V_b$, resulting in a much steeper dependence $M_p(V)$ than we have in our case[7]. It is worth noting that the same type of asymmetry, $\beta \gg \alpha$, was earlier reported for 6H SiC[1,4].

The coefficient of hole-initiated multiplication is given by the relationship[6,7]:

$$1 - \frac{1}{M_p} = \int_0^W \beta \exp\left(- \int_W^x (\beta - \alpha)\, dx' \right) dx. \tag{1}$$

Here the metallurgical p-n junction position is supposed to be at $x = W$. The expression for M_n is symmetrical to Eq. (1). Extraction of ionization coefficients from photomultiplication data was done by fitting simulated $M(V)$ dependencies to experimental curves. This procedure is similar to the ''composite sample technique''[7]. The simulated dependencies are obtained by numerical calculation of the ionization integral in Eq. (1) using the exact doping profiles as obtained from C-V measurements.

As the first step we establish an approximate dependence of β and α on the electric field using simplified relationships for electron and hole ionization rates. At this stage individual fits to multiplication curves were obtained for each sample using the approximation of proportional ionization coefficients $K = \beta/\alpha = \mathrm{const}$ and exponential dependence of ionization rates on reciprocal electric field: $\beta = \beta_0 \exp(-E_0/E)$. Here E is the electric field and E_0 is a characteristic field. The β/α ratio is assumed to be independent of electric field at this stage, a constant for each sample. The fitting procedure is illustrated by Fig. 2(b). M_p is represented as two functions: $f_1 = 1 - M_p^{-1}$ and $f_2 = M_p^{-1}$. The first function, f_1, is almost insensitive to the value of K. It is governed by the slope of the field dependence of β, i.e., by E_0. The plot of calculated $1/M_p$ assuming $K = \infty$ ($\alpha = 0$) is also shown for comparison. It is seen that the avalanche multiplication is dominated by holes, the contribution of electrons is only significant for the region very close to the avalanche breakdown. Once the fit to experimental f_1 is obtained, the value of K can be optimized so as to have a have a good fit to f_2.

The second step of the procedure is fitting to experimental photomultiplication curves using one and the same relationship for $\alpha(E)$ and $\beta(E)$ for all the samples . The theoretical dependence for hole ionization rates was used in the form proposed by Thornber[7,8]:

$$\beta_{th} = \frac{qE}{\mathcal{E}_i^h} \exp\left(- \frac{\mathcal{E}_i^h}{(qE\lambda_h)^2 / + qE\lambda_h} \right), \tag{2}$$

where λ_h is the mean free path, $\mathcal{E}_i^h$ is the ionization energy, and $\mathcal{E}_r$ the optical phonon energy (0.12 eV for SiC). For electrons a modified form of Eq. (2) was used, which retains only the high-field asympthotics of Eq. (2):

$$\alpha_{th} = \frac{qE}{\mathcal{E}_i^e} \exp\left(- \frac{3\mathcal{E}_i^e \mathcal{E}_r}{(qE\lambda_e)^2} \right), \tag{3}$$

where $\mathcal{E}_i^e$ and λ_e are the corresponding parameters for electrons. The fitting procedure was nearly the same as that used in the first step. The ionization coefficients used to fit the data were $\alpha = k_\alpha \alpha_{th}$ and $\beta = k_\beta \beta_{th}$. The data used in the fittings are plotted in Fig. 3 together with the theoretical dependencies. The photomultiplication coefficients calculated using these values of β and α are plotted in Fig. 2(a).

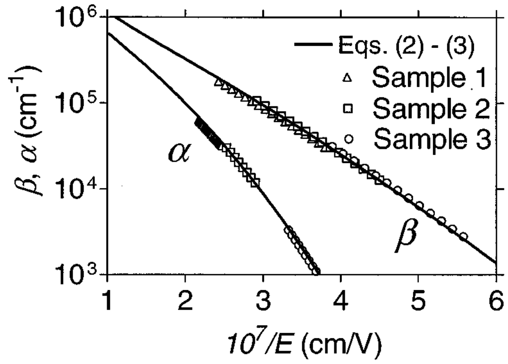

FIG. 3. Impact ionization rates for electrons and holes in 4H SiC. Points show the results of fitting to experimental data, continuous lines represent the theoretical model employed.

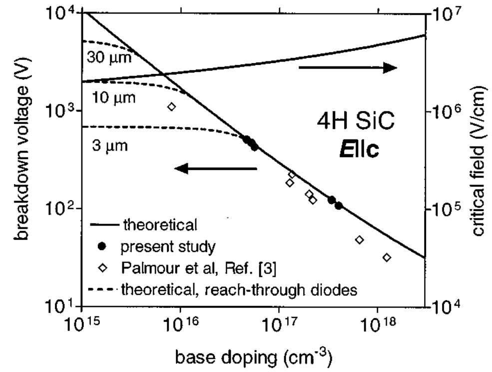

FIG. 4. Dependence of breakdown voltages and critical fields in 4H SiC junction devices on base doping. Dashed lines correspond to reach through diodes with a base thickness of 3, 10 and 30 μm.

The following values of the parameters were obtained for the theoretical model: $\mathscr{E}_i^h$ =7 eV, $\mathscr{E}_i^e$ = 10 eV, λ_h=32.5 Å, λ_e=29.9 Å, and $\mathscr{E}_r$ =0.12 eV.

The strong suppression of impact ionization by electrons we observe for 4H SiC may seem quite paradoxical. Electrons in 4H SiC have indeed a lower effective mass than holes and much higher mobility. On the other hand, the same type of ionization asymmetry was earlier observed for 6H. That was attributed to the discontinuity of the electron energy spectrum for motion along the principal axis[1]. Recent band structure calculations have shown the existence of the same type of discontinuity for 4H SiC.[9,10]. The low-field solution of the Boltzmann equation $\alpha \propto \exp(-E_\alpha/E)$ physically corresponds to Shockley's ''lucky electron'' model, which implies a ballistic type of ionization process[7]. This process is forbidden for the discontinuous energy spectrum such as in 4H SiC, since an electron cannot reach the ionization threshold without either phonon collisions or tunneling. The low-field Shockley solution of the Boltzmann equation no longer holds, which is the reason why we have omitted the corresponding term in Eq. (3). Another reason for the suppression of impact ionization by electrons comes from the high energy losses of hot carriers due to Bragg reflections from the zone edges (the Bloch oscillations), as was earlier suggested for 6H SiC[1]. Recent numerical simulations of electron drift velocities in 4H SiC indeed confirm the importance of Bloch oscillations for the high field electron transport along the c axis[10]. The valence band in SiC has a continuous energy spectrum, which is essentially the reason why impact ionization by holes is much more efficient.

One immediate application of experimental data on ionization rates is determination of theoretical critical fields and breakdown voltages in p-n junction devices. Using Eqs. (1)–(3) we have numerically calculated the breakdown voltages and fields for one-sided abrupt p^+-n junctions in 4H SiC. The results of the calculation are shown in Fig. 4. Experimental data plotted in Fig. 4 are the breakdown voltages of sample Nos. 2–3 and of a few more diodes from the same wafers. Results of Palmour *et al.*[3] are also shown for comparison. For the high voltage/low doping range, $V_b > 300$ V the following power-law approximation can be used for the theoretical dependence: $V_b = 1720(N/10^{16}\mathrm{cm}^{-3})^{0.8}$ (V), where N is the base doping. At a higher doping level the dependence has a lower exponent because of a weaker dependence of ionization rates on the electric field. Also plotted in Fig. 4 are the theoretical breakdown voltages for reach through diodes with a base thickness of 3,10, and 30 μm. The critical field in 4H SiC can be approximated by the relationship:

$$E_{cr} = \frac{2.49\times 10^6}{1-\frac{1}{4}\log_{10}(N/10^{16}\mathrm{cm}^{-3})}\ \mathrm{V/cm}. \tag{4}$$

The critical field in 4H SiC appears to be more than 10 times higher than in silicon.

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